

Folding from predicted contacts instead of MSAs

Helico + MarinerFold · results on 98 FoldBench monomers

github.com/Open-Athena/helico · PR #13, issue #11

The idea

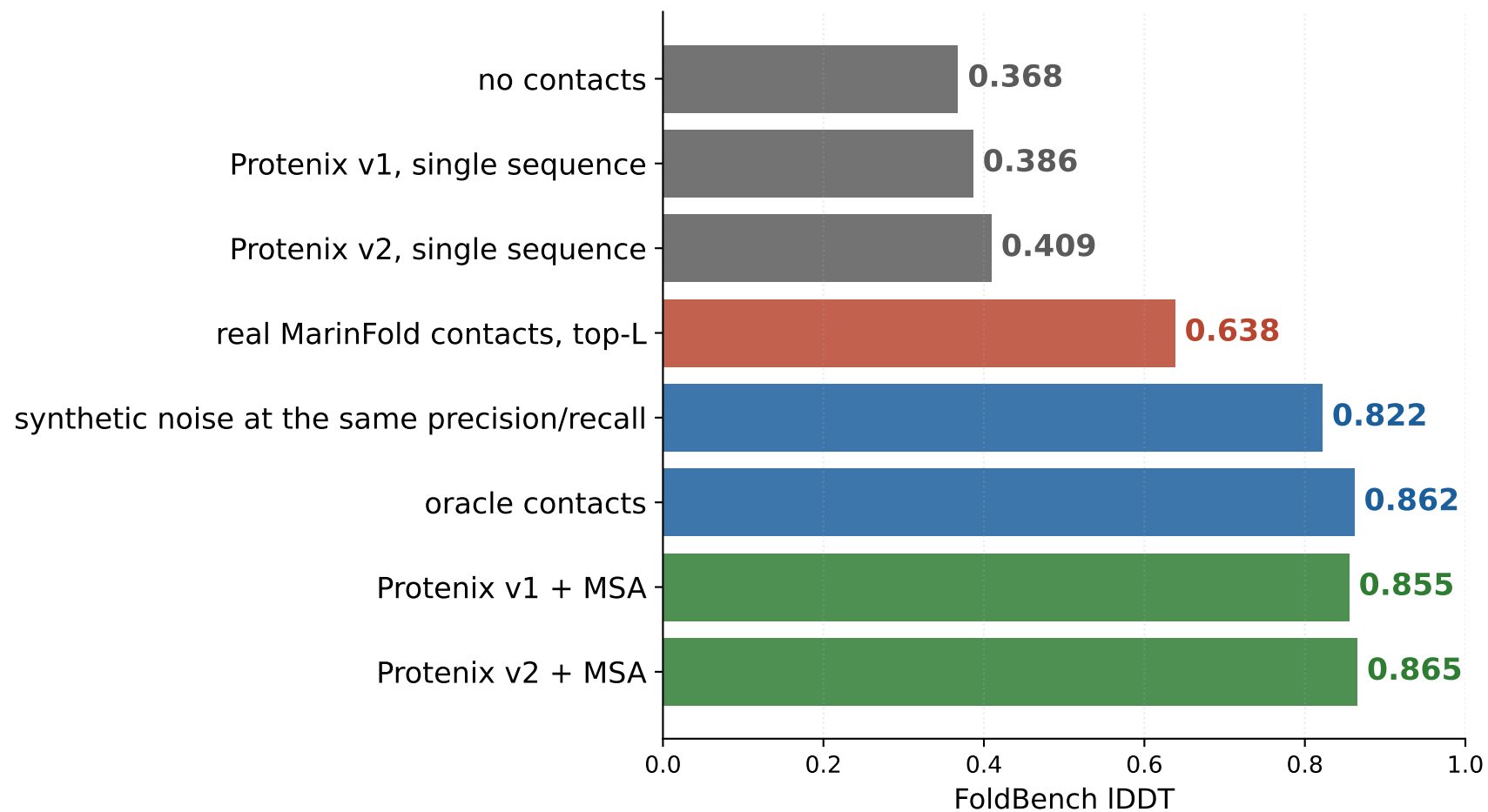
AF3-family models lean on MSAs. Strip the alignment and accuracy collapses.

- MarFold predicts residue-residue side-chain contacts directly.
- Feed those contacts to the folding model in place of the alignment, and the MSA search leaves the critical path.
- Helico takes a three-state contact matrix — contact / no-contact / unknown — added into the pair representation.
- Training samples the conditioning level per example, so one model spans no contacts through fully specified.

All Helico arms here are genuinely MSA-free: no alignment, no conservation profile, at training or inference.

Where things land

FoldBench, 91 paired monomer targets, all MSA-free



**Real contacts beat
single sequence
by a wide margin**

**...but do not reach
MSA-level accuracy**

Yes: real contacts clearly beat single-sequence folding

vs Protenix v2 -- the stronger baseline -- in single-sequence mode

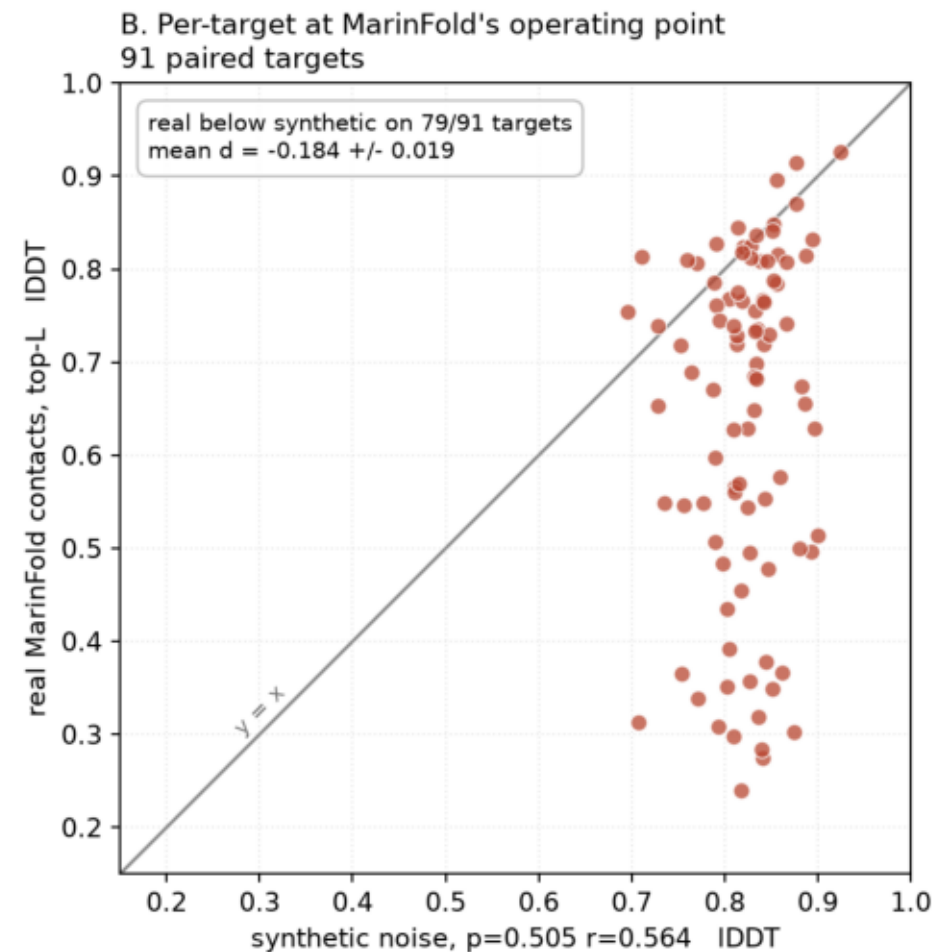
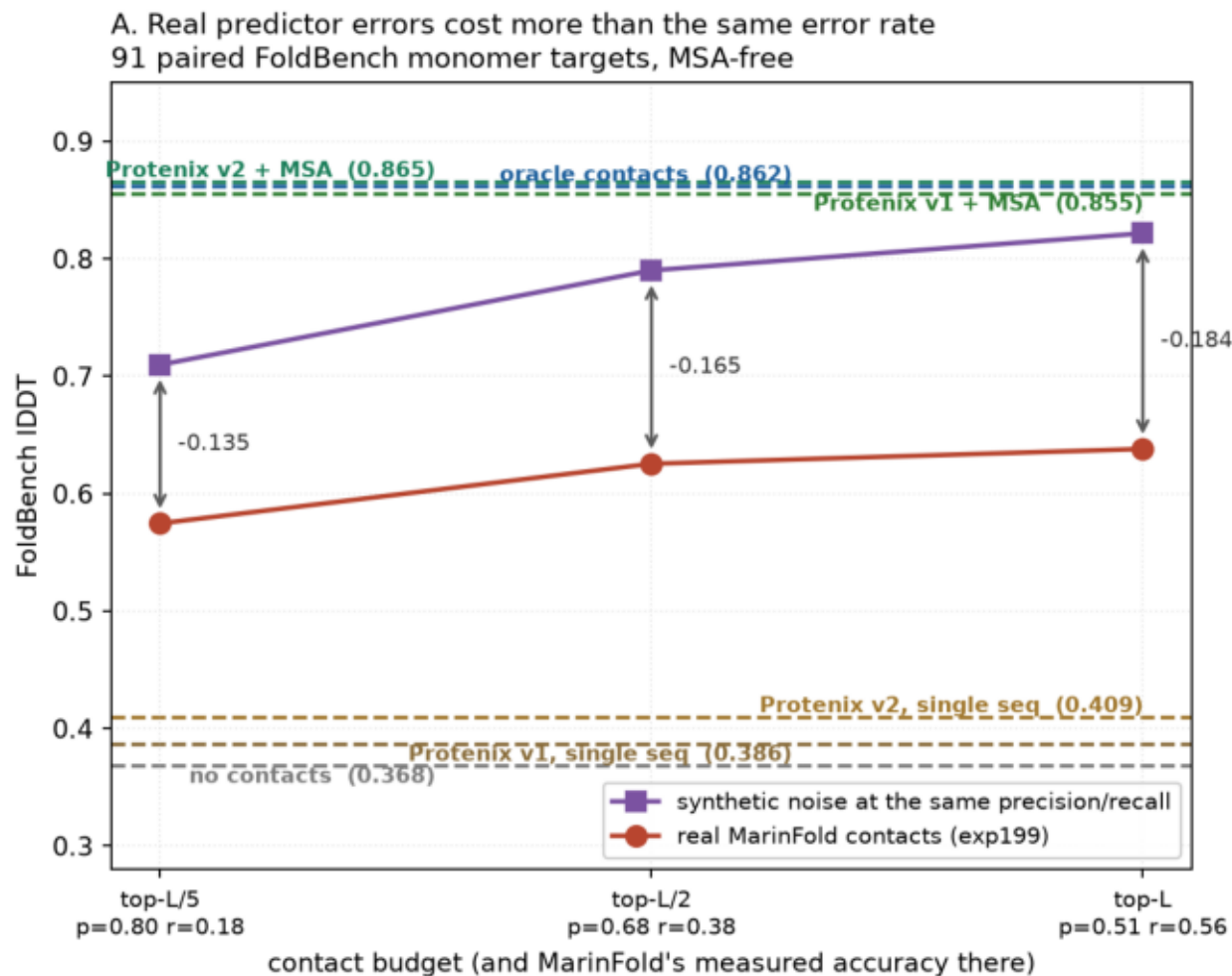
- Real MarFold contacts (top-L): $+0.229 \pm 0.022$ IDDT (t=10.4, better on 76/91)
- Protenix v2 is genuinely the harder baseline: +0.023 over v1 without MSAs, +0.010 with them. The margin survives anyway.
- Holds at every contact budget, and even a single rollout beats it.

The control that makes this credible

Our own contacts-off arm scores 0.368 — BELOW Protenix v2's 0.409 (-0.042 ± 0.008). The fine-tuned model has no intrinsic edge with no information. All of the gain comes from the contacts.

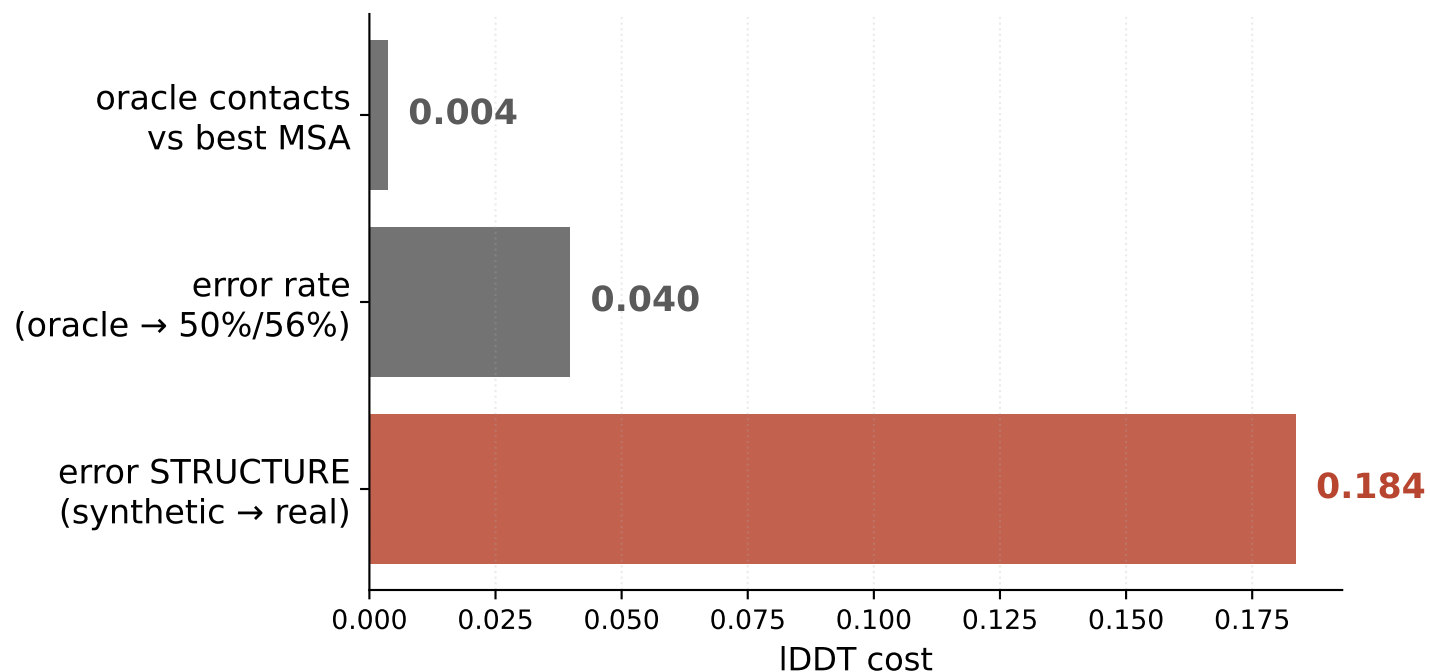
But error structure, not error rate, is what hurts

each synthetic arm generated at the precision/recall measured for its real counterpart



Decomposing the 0.227 gap to MSAs

real MarFold top-L 0.638 → Protenix v2 + MSA 0.865



Real predictor errors cluster near true contacts, where they are geometrically plausible and cannot be rejected as inconsistent with the rest of the map. Our training noise model draws false positives uniformly — the easy case.

Oracle contacts vs the best MSA model is +0.004 — indistinguishable. A perfect

What to do about it

- Retrain with false positives sampled from near-miss pairs rather than uniformly. The model has never seen the error distribution it faces. Cheapest lever, and it is on our side rather than MarFold's.
- Vote aggregation across rollouts is worth $+0.071 \pm 0.008$ over a single rollout at the same budget — keep using it, never mix the recipes.
- Improve MarFold's error profile, not only its precision/recall.

Caveats

Warm-started from Protenix v1. Monomers only, so not comparable to the earlier assembly-set numbers. MarFold indexes into the published prompt, Helico into resolved residues — only 15/100 agree outright; the map is verified by round trip at Jaccard 0.998.